

AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Project Documentation

ServiceNow Flow Designer Based Automation Project

1. Abstract

Auto Ticket Classification using Flow Designer is an automation-based project designed to classify and route service tickets automatically. In a traditional ticket management system, support teams manually read each ticket, identify its category and priority, and assign it to the appropriate department or team. This process can be time-consuming and may result in delays or incorrect assignments.

The proposed system uses ServiceNow Flow Designer to automate the ticket classification process. When a new ticket is created, the system checks ticket information such as description, category, priority, and other conditions. Based on predefined rules, the ticket is automatically classified and assigned to the appropriate group.

This system reduces manual effort, improves ticket handling speed, provides consistent classification, and helps support teams manage incidents more efficiently.

2. Introduction

Ticket management is an important part of IT service management. Organizations receive different types of service requests and incidents every day. These tickets need to be categorized, prioritized, and assigned to the correct support team.

Manual ticket classification requires human involvement for every new ticket. This can increase workload and may cause delays.

Flow Designer provides a visual, low-code environment for creating automated workflows. Using Flow Designer, business rules and actions can be connected to create an automatic ticket classification process.

3. Problem Statement

In a manual ticket management system, support staff need to examine every ticket and decide its category, priority, and assignment group. This can lead to increased manual workload, delayed ticket assignment, incorrect classification, human errors, slow response to important incidents, and difficulty in handling a large number of tickets.

Therefore, an automated system is required to classify and route tickets quickly and consistently.

4. Objectives

- Automate ticket classification.
- Reduce manual effort in ticket management.
- Automatically identify the appropriate ticket category.
- Assign tickets to the correct support group.
- Set ticket priority based on predefined conditions.
- Reduce ticket processing time.

- Improve the overall efficiency of IT service management.
- Provide consistent ticket handling.

5. Existing System

In the existing system, tickets are generally processed manually.

Process:

1. User creates a ticket.
2. Support staff opens the ticket.
3. Staff reads the ticket description.
4. Ticket category is identified manually.
5. Priority is selected.
6. Ticket is assigned to a support group.
7. Support team starts processing the ticket.

Disadvantages:

- Time-consuming
- Requires manual effort
- Possibility of human errors
- Delayed assignment
- Difficult to manage high ticket volume
- Inconsistent classification

6. Proposed System

The proposed system uses ServiceNow Flow Designer to automate the ticket classification process. When a ticket is created, a flow is automatically triggered. The flow checks predefined conditions and performs appropriate actions.

Automated Process:

New Ticket → Trigger Flow → Check Ticket Details → Classify Ticket → Set Priority → Assign Group → Update Ticket → Notify Team

7. System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection
- Standard processor

Software Requirements

- ServiceNow Platform

- ServiceNow Flow Designer
- Web Browser
- Windows/Linux/macOS
- Internet connection

8. Technologies Used

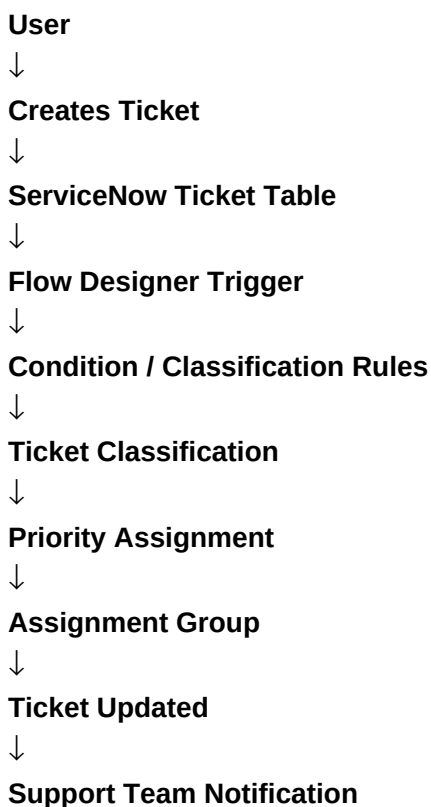
ServiceNow: A cloud-based platform used for IT service management and business process automation.

Flow Designer: A visual automation tool in ServiceNow that allows users to create workflows using triggers, conditions, actions, and subflows with minimal coding.

ServiceNow Tables: The project can use the Incident table or another suitable ticket table to store and manage ticket information.

9. System Architecture

The basic architecture of the system is:



10. Modules

Module 1: Ticket Creation

The user creates a new ticket by entering details such as short description, description, category, impact, urgency, and caller information.

Module 2: Flow Trigger

Flow Designer monitors the selected ticket table. When a new ticket satisfies the trigger condition, the flow starts automatically.

Module 3: Ticket Classification

The flow checks ticket information using predefined conditions and identifies issues such as Hardware, Software, Network, Account, or Access issues.

Module 4: Priority Assignment

The flow determines ticket priority based on impact and urgency.

Module 5: Assignment

After classification, the ticket is automatically assigned to the appropriate support group.

Module 6: Notification

A notification can be sent to the responsible support team after classification and assignment.

11. Flow Designer Workflow

Step 1: Create a Flow

Create a new flow in ServiceNow Flow Designer.

Step 2: Configure Trigger

Select the ticket or Incident table. Example: Record Created.

Step 3: Add Conditions

The flow checks ticket information. For example, a network-related description can be classified as a Network Issue.

Step 4: Set Category

Update the category field according to the classification rule.

Step 5: Set Priority

Determine priority using predefined conditions.

Step 6: Assign Group

Assign the ticket to the appropriate support group.

Step 7: Update Ticket

Update the ticket with classification and assignment details.

Step 8: Send Notification

Send a notification to the assigned support team.

12. Sample Classification Logic

Condition	Classification	Assignment
Network connectivity problem	Network	Network Support
Computer hardware problem	Hardware	Hardware Support
Application/software problem	Software	Software Support
Password/account problem	Account	IT Support
Unauthorized access problem	Security	Security Support

13. Example

Input Ticket: Unable to connect to the office Wi-Fi.

Flow Processing: The Flow Designer checks the ticket description and identifies the issue as a network-related problem.

Automatically Updated Information:

Category: Network

Priority: Based on impact and urgency

Assignment Group: Network Support

Status: Assigned

14. Advantages

- Reduces manual work
- Faster ticket classification
- Faster assignment
- Reduces human errors
- Provides consistent processing
- Improves support team productivity
- Helps prioritize important tickets
- Can handle large numbers of tickets
- Easy to modify and maintain
- Provides better ticket management

15. Applications

- IT help desks
- Educational institutions
- Software companies
- Banking organizations
- Healthcare organizations
- E-commerce companies

- Customer support centers
- Corporate IT departments

16. Limitations

- Classification depends on configured rules.
- Incorrect or incomplete ticket information may affect classification.
- Complex classification may require additional automation or scripting.
- The system depends on the availability of the ServiceNow platform.
- Rules need to be maintained when organizational requirements change.

17. Future Enhancements

1. AI-based ticket classification.
2. Natural Language Processing (NLP).
3. Automatic sentiment analysis.
4. Machine-learning-based priority prediction.
5. Automatic duplicate ticket detection.
6. Intelligent assignment based on technician workload.
7. Predictive ticket routing.
8. Automatic response generation.
9. Performance analytics and dashboards.
10. Integration with email and messaging platforms.

18. Expected Outcome

The proposed system is expected to automatically classify incoming tickets and route them to the appropriate support group. It helps reduce the time required for manual ticket processing and improves the efficiency of the support team. The system provides a structured and automated approach to ticket management.

19. Conclusion

Auto Ticket Classification using Flow Designer provides an efficient method for automating ticket management in ServiceNow. The system automatically processes newly created tickets, classifies them according to predefined conditions, assigns appropriate priority, and routes them to the correct support group.

By reducing manual intervention, the system helps improve processing speed, consistency, and productivity. Flow Designer also provides a visual and low-code approach, making the automation easier to develop and maintain.

Overall, the project demonstrates how workflow automation can improve IT service management and

provide faster handling of service tickets.

20. References

1. ServiceNow Documentation
2. ServiceNow Flow Designer Documentation
3. ServiceNow IT Service Management Documentation
4. ServiceNow Developer Resources
5. IT Service Management concepts and workflow automation references